

# Nguyen Anh Khoi

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## Professional Summary

**Java Backend Developer** focused on building secure REST APIs, transaction-safe workflows, and event-driven services with Spring Boot. Hands-on experience with Spring Security, JPA, SQL databases, Redis, RabbitMQ, gRPC, and Docker, with measurable improvements in query latency, processing time, and concurrent workload capacity.

## Technical Skills

**Languages:** Java, SQL

**Java Backend:** Spring Boot 3, Spring Security, Spring Data JPA, Hibernate/JPA

**Data & Messaging:** SQL Server, PostgreSQL, Redis, RabbitMQ

**Backend Engineering:** REST APIs, JWT/RBAC, Transaction Management, Query Optimization, gRPC, WebSocket

**Tools:** Maven, Flyway, Docker, Docker Compose, Git, MinIO, Jsoup

## Experience

### THD Cybersecurity

Jan 2026 – Jun 2026

*Backend Developer Intern / EMIS Voice Detection*

*Ho Chi Minh City, Vietnam*

- Developed **13 Spring Boot backend modules** for meeting lifecycle, target management, voice profiles, audio samples, chatbot, and conversation history within a **21-service** AI meeting intelligence platform.
- Designed RabbitMQ-based scheduling and worker orchestration for automated meeting recording, scaling bot capacity from **2 to 10 concurrent sessions**.
- Reduced audio handoff latency from **3 minutes to 15 seconds** by optimizing upload callbacks, backend processing, and downstream service coordination.
- Implemented **gRPC streaming**, WebSocket event delivery, Redis-backed state management, MinIO audio storage, JWT/2FA, and secure service-to-service communication.

**Technologies:** *Java 17, Spring Boot 3, Spring Security, Spring Data JPA, PostgreSQL, Redis, RabbitMQ, gRPC, WebSocket, MinIO, Docker*

## Projects

### Carbon Credit Marketplace

Sep 2025 – Nov 2025

*Backend Developer / GitHub Repository*

- Owned backend delivery for a **Spring Boot 3** carbon credit marketplace, implementing **120 REST endpoints** across **18 domain modules** and **18 SQL Server entities**.
- Designed transaction-safe purchase and transfer workflows using row-level locking, idempotent completion, compensating actions, and audit logs to preserve wallet and carbon-credit ownership consistency.
- Secured protected endpoints with Spring Security, stateless JWT, token blacklisting, and method-level **RBAC across 4 roles**; integrated validated VNPAY and MoMo payment callbacks.
- Reduced query latency from **1,437 ms to 24 ms** by optimizing data-access paths and adding Flyway-managed indexes and constraints; enforced a **30 requests/minute** limit on sensitive endpoints.

**Technologies:** *Java 17, Spring Boot 3, Spring Security, Spring Data JPA, SQL Server, Flyway, JWT, Maven, VNPAY, MoMo*

### Premier League Stats

Apr 2025 – Jul 2025

*Backend Developer*

- Built a **Spring Boot 3** analytics backend and Java-based Jsoup ingestion pipeline that normalized fixtures, standings, and match statistics into **13 entities** and exposed **59 REST endpoints**.
- Improved crawler resilience through proxy routing, request throttling, retry handling, and record deduplication across repeated ingestion jobs.
- Reduced ingestion time from **30 seconds to 4 seconds** and memory usage from **17 MB to 493 KB** by separating fetch, parse, transform, and persistence stages while minimizing retained objects.
- Developed database-backed match analytics and formula-based prediction features exposed through Spring Security-protected, role-based REST APIs.

**Technologies:** *Java 17, Spring Boot 3, Spring Security, Spring Data JPA, SQL Server, Jsoup, JWT, Maven*

## Education

### FPT University

Sep 2023 – Expected Oct 2027

*Bachelor of Software Engineering*

*GPA: 3.2/4.0*

## Certificate

Oracle Cloud Infrastructure Certified AI Foundations Associate

August - 2026